

26.     **Ans: D**

$$E_2 - E_1 = \frac{hc}{\lambda_2} = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{1.78 \times 10^{-7}}$$
$$= 1.12 \times 10^{-18} \text{ J}$$

$$E_3 - E_2 = \frac{hc}{\lambda_1} = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{6.22 \times 10^{-7}}$$
$$= 0.32 \times 10^{-18} \text{ J}$$

$$\text{Hence } E_3 - E_1 = (1.12 + 0.32) \times 10^{-18}$$
$$= 1.44 \times 10^{-18} \text{ J}$$
$$= 9.00 \text{ eV}$$

Hence the minimum p.d to excite electrons must be 9.00 V.

27.     **Ans: D**